



## Bank Marketing

Donated on 2/13/2012

The data is related with direct marketing campaigns (phone calls) of a Portuguese banking institution. The classification goal is to predict if the client will subscribe a term deposit...



### Dataset Characteristics

Multivariate

### Subject Area

Business

### Associated Tasks

Classification

### Feature Type

Categorical, Integer

### # Instances

45211

### # Features

16

## Dataset Information



### Additional Information

The data is related with direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, in order to access if the product (bank term deposit) would be ...

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### Has Missing Values?

No

## Introductory Paper



### [A data-driven approach to predict the success of bank telemarketing](#)

By Sérgio Moro, P. Cortez, P. Rita. 2014

Published in Decision Support Systems



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| Variable Name | Role    | Type        | Demographic     | Description                                                                                                                                                                                                                                                                               |
|---------------|---------|-------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| age           | Feature | Integer     | Age             |                                                                                                                                                                                                                                                                                           |
| job           | Feature | Categorical | Occupation      | type of job (categorical: 'admin.','blue-collar','employed','services','student','technician')                                                                                                                                                                                            |
| marital       | Feature | Categorical | Marital Status  | marital status (categorical: 'divorced','married','widowed')                                                                                                                                                                                                                              |
| education     | Feature | Categorical | Education Level | (categorical: 'basic.4y','basic.6y','basic.9y','high.school')                                                                                                                                                                                                                             |
| default       | Feature | Binary      |                 | has credit in default?                                                                                                                                                                                                                                                                    |
| balance       | Feature | Integer     |                 | average yearly balance                                                                                                                                                                                                                                                                    |
| housing       | Feature | Binary      |                 | has housing loan?                                                                                                                                                                                                                                                                         |
| loan          | Feature | Binary      |                 | has personal loan?                                                                                                                                                                                                                                                                        |
| contact       | Feature | Categorical |                 | contact communication type (categorical)                                                                                                                                                                                                                                                  |
| day_of_week   | Feature | Date        |                 | last contact day of the week                                                                                                                                                                                                                                                              |
| month         | Feature | Date        |                 | last contact month of year (categorical)                                                                                                                                                                                                                                                  |
| duration      | Feature | Integer     |                 | last contact duration, in seconds (numeric). Target variable (y) is binary classification: 1 = success (e.g., if duration=0 then y='no'). Y=0 means client was not contacted. Also, after the end of the call y is obvious for benchmark purposes and should be discarded from the model. |
| campaign      | Feature | Integer     |                 | number of contacts performed during this marketing campaign                                                                                                                                                                                                                               |
| pdays         | Feature | Integer     |                 | number of days that passed by after the last contact (numeric; -1 means client was not previously contacted)                                                                                                                                                                              |
| previous      | Feature | Integer     |                 | number of contacts performed before this marketing campaign                                                                                                                                                                                                                               |
| outcome       | Feature | Categorical |                 | outcome of the previous marketing campaign                                                                                                                                                                                                                                                |



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## Additional Variable Information



Input variables:

# bank client data:

1 - age (numeric)

2 - job : type of job (categorical:

"admin.", "unknown", "unemployed", "management", "housemaid", "entrepreneur", "student",  
"blue-collar", "self-employed", "retired", "technician", "services")

3 - marital : marital status (categorical: "married", "divorced", "single"; note: "divorced" means divorced or widowed)

4 - education (categorical: "unknown", "secondary", "primary", "tertiary")

5 - default: has credit in default? (binary: "yes", "no")

6 - balance: average yearly balance, in euros (numeric)

7 - housing: has housing loan? (binary: "yes", "no")

8 - loan: has personal loan? (binary: "yes", "no")

# related with the last contact of the current campaign:

9 - contact: contact communication type (categorical: "unknown", "telephone", "cellular")

10 - day: last contact day of the month (numeric)

11 - month: last contact month of year (categorical: "jan", "feb", "mar", ..., "nov", "dec")

12 - duration: last contact duration, in seconds (numeric)

# other attributes:

13 - campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)

14 - pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric, -1 means client was not previously contacted)

15 - previous: number of contacts performed before this campaign and for this client (numeric)

16 - poutcome: outcome of the previous marketing campaign (categorical: "unknown", "other", "failure", "success")

Output variable (desired target):

17 - y - has the client subscribed a term deposit? (binary: "yes", "no")

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| File                | Size     |
|---------------------|----------|
| bank.zip            | 565.5 KB |
| bank-additional.zip | 434.2 KB |

## Papers Citing this Dataset ^



SORT BY YEAR, DESC

### [Fair Algorithms for Clustering](#)

By Suman Bera, Deeparnab Chakrabarty, Nicolas Flores, Maryam Negahbani. 2019  
Published in ArXiv.

### [Clustering with Fairness Constraints: A Flexible and Scalable Approach](#)

By Imtiaz Ziko, Eric Granger, Jing Yuan, Ismail Ayed. 2019  
Published in ArXiv.

### [Noise-tolerant fair classification](#)

By Alexandre Lamy, Ziyuan Zhong, Aditya Menon, Nakul Verma. 2019  
Published in ArXiv.

### [AdaFair: Cumulative Fairness Adaptive Boosting](#)

By Vasileios Iosifidis, Eirini Ntoutsi. 2019  
Published in

### [Quantification under prior probability shift: the ratio estimator and its extensions](#)

By Afonso Vaz, Rafael Izbicki, Rafael Stern. 2018  
Published in ArXiv.

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## DOI

[10.24432/C5K306](https://doi.org/10.24432/C5K306)

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